PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Hypertensive Disorders in Pregnancy and Risk of Asthma in Offspring: A Systematic Review and Meta-Analysis
AUTHORS	Li, Ping; Xiong, ta; Hu, Yong

VERSION 1 – REVIEW

REVIEWER	Malhamé, Isabelle McGill University
REVIEW RETURNED	26-Jan-2021

GENERAL COMMENTS	This is an interesting topic, and a methodologically sound systematic review. In order to improve the clarity of the manuscript, I would have the following comments for the investigators: (Page numbers referred to are those marked at the bottom of each page, and line numbers are those at the extreme left of each page) Page 4: strength and limitations: Remove the word finally on line 8 Introduction Page 6: HDP should be defined according to the International Society of Hypertension in Pregnancy. Page 7: The goal of the systematic review should be reframed. This systematic review will not help the investigators to establish a causal link between HDP and childhood asthma, but rather to clarify the magnitude of this association. Methods: Page 8, line 56: How were case series and case reports handled? Page 9, line 18: Clarify what is meant by “studies with overlapping data” and “studies without raw data”. Page 9, line 36: Do you mean that studies were retrieved from each “article”? (currently written as "database") Page 10, line15: How did you score the other study designs? Page 10, line 56: Were all subgroup analyses defined a priori? Were some defined a

	posteriori? Please clarify this, as well as the rationale for the subgroup analyses. In particular, the rationale for the sibling analysis should be explained here and not in the results section. There is a section missing on the description of how the authors assessed what were “appropriate confounders” adjusted for in each individual study. I defer to a statistical expert regarding the appropriateness of the analysis used. Results: Clear description of the results. Discussion: Page 19: The discussion should start with a brief summary of the study findings, without repeating the results. In addition, the overall tone of the discussion should be brought down, as the association found was quite moderate, and I suspect that many confounders were not captured by individual studies. Elements currently included in paragraph 1 of the discussion should be featured in the investigator’s perceived strengths and weaknesses section, and do not belong to the first paragraph of the discussion. Regardless of the causal link between HDP and asthma, HDP appears to be an interesting predictor of asthma in the offspring. Hence, a paragraph in the discussion should expand further on what “early surveillance and interventions” would be warranted for infants of mothers of HDP. Perhaps, assessing whether maternal HDP was present when evaluating infants for asthma would be important.

REVIEWER	Abdullah, Kawsari Children's Hospital of Eastern Ontario, Research Institute
REVIEW RETURNED	29-Jan-2021

GENERAL COMMENTS	This manuscript aimed to summarize the evidence showing an association between hypertensive disorders in pregnancy (HDP) and the risk of asthma in offspring through performance of a systematic review and pooling of the data from 10 cohort studies using meta-analysis technique. Discerning the perinatal origin of chronic diseases such as asthma can play an important factor in its early detection and management. The authors have done a commendable job in describing the rationale of this study. Authors have used the PRISMA and MOOSE guidelines in the reporting of this systematic review with meta-analysis. Despite their use of standard reporting guidelines there are specific sections in the review that were quite unorganized and lacked clarity. The quality of the writing needs to be further improved both in terms of grammar and meaning. However, my comments below are purely focused on the methodological and scientific merits of the manuscript. Introduction: 1. On page 5 line#59-62 authors state some examples of offspring diseases that have been shown to have association with HDP. Here they do not mention respiratory diseases. Mentioning respiratory conditions here can provide a good segue to the next sections where they discuss the association with respiratory diseases in

	detail. Methods: 1. Page 7 line#91: The “data source and search strategy” title heads the wrong paragraph. This paragraph contains information related to the “Eligibility criteria” of the studies. Additionally, these criteria are not complete and lack clarity. Certain information related to the outcome have been provided in the section titled “study selection”. Authors have stated that they have used the PRISMA checklist for developing the chronology and content of the methods section. However, when I compare the section heading and content provided within these headings to the PRISMA checklist, there are inconsistency and absence of information. One example has been mentioned in my comment above. Hence, I find the methods section of this review messy and lacking the systematic approach of a review. 2. Some secondary outcomes have been stated on page 7 line#95, these include active asthma and asthma exacerbation. Why are these two not considered as asthma? How were they defined? 3. On page 9 line#119: What does “literature that was peer reviewed” mean? If the authors only included studies that were published after peer review, it is better to refer to it as “peer reviewed published study” instead of the above. 4. Page 10 line#143-144, not clear what authors are meaning by stating - “hazard ratios, incidence rates and RRs were directly considered ORs in the pooled analysis. These effect estimates provide different estimates of risk association. Hence, further clarification on how they were pooled is needed. Results: 1. On page 14 line# 180: What does 252,450 cases mean? How are cases defined? 2. Page 15 line#201: Authors have shown a pooled estimate of the adjusted OR (1.19,95%CI 1.12 – 1.26). However, it is also shown that this estimate had statistically significant heterogeneity (I2 = 72.5%). With this high level of heterogeneity should the authors report the pooled value as their primary estimate of the association between HDP and offspring asthma as depicted in the abstract, results and discussion section of the review. 3. Subgroup analysis results: Within each of the subgroup analysis (table 1), the ORs shown as statistically significant, also show high level of heterogeneity (category of HDP= 77%; sibling design (no) = 75.8%; study quality = 66.9%; location = 78.5%; exposure assessment = 70.1%; outcome assessment = 82.8%). Based on this observation, the question again arises whether pooling of the effect estimates (ORs) reported in each of the selected studies is justifiable? 4. Subgroup analysis results: It has been stated on page 18 line#235 that after adjusting for maternal smoking during pregnancy the heterogeneity has decreased. However, in table 1 the heterogeneity reported without adjustment for this variable is less than the adjusted one (adjusted (30.7) vs. not adjusted (29.9)). Please verify why? 5. Formatting of table 2:

	□ What does the column heading “heterogeneity statistic” imply? □ Having the I2 column before the OR is confusing. 6. Supplemental Table 4: Does each row estimate indicate the pooled ORs after excluding that particular study? If so, a heading or a footnote is needed. Otherwise, the results given in this table are not interpretable just on own its own. Discussion and conclusion: 1. Page 20 line#278: there should not be a dash after two. 2. Page 21 line#292: Authors have stated that misclassification due to different assessment of HDP and Outcomes. From table 1 it is evident that many of the studies used ICD codes for ascertainment of HDP. A subgroup analysis has been done based on exposure assessment where the categories were maternal-report and medical records. Should this category be clinical diagnosis of HDP vs. ICD codes based medical records? Clinical method is the gold standard for any disease diagnosis and able to provide a more robust evidence associated with any risk association. This fact may need to be highlighted in the conclusion where the authors are calling for further research.

REVIEWER	Liu, Zhengping Southern Medical University, Department of Obsterics
REVIEW RETURNED	01-Apr-2021

GENERAL COMMENTS	Thank you for giving me the opportunity to review this manuscript. I have following concerns based on the statistical method. 1. Complete the result of records obtained by each search strategy in Supplemental eTable 1. 2. The authors should explain the rationale for the statistical method of combining ORs for subtypes of HDPs. The definitions of exposures (GH, PE and HbP) are very different though they are subtypes of HDPs. Otherwise, the meta-analysis should be performed by subtypes separately. 3. For the study by Byberg et al., the ORs for MM-PE and S-PE were derived from the same logistic model, in which no-PE was the shared reference. Therefore, directly combining these ORs in the meta-analysis is problematic since these ORs are not independent. Alternatively, a fixed effects model is suggested to calculate a combined OR for PE and this combined OR can be used for meta-analysis afterwards. [1] [1] Dong JY, Zhang YH, Qin LQ. Erectile dysfunction and risk of cardiovascular disease: meta-analysis of prospective cohort studies. J Am Coll Cardiol. 2011 Sep 20;58(13):1378-85. doi: 10.1016/j.jacc.2011.06.024. PMID: 21920268.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Dr. Isabelle Malhamé, McGill University

Page 4: strength and limitations:
Remove the word finally on line 8
Response: Dear Dr. Isabelle Malhamé, thank you very much for your thorough review. We removed the word“finally”from line 8. Please see line 52 on page 4.

Introduction
Page 6:
HDP should be defined according to the International Society of Hypertension in Pregnancy.
Response: Thank you for this valuable question. We agree. Although there are several HDP definitions, these definitions were very similar across different studies. In the revised manuscript, HDP were defined according to the International Society of Hypertension in Pregnancy. HDP were defined as any hypertension condition (systolic blood pressure (BP) ≥140 and/or diastolic BP ≥90 mmHg) that manifested before gestation or before 20 weeks and hypertension starting at or after 20 weeks according to current guidelines, which also included the International Society for the Study of Hypertension in Pregnancy guidelines. We added the definition and a reference in the introduction. Please see lines 56~59 on page 5.

Page 7:
The goal of the systematic review should be reframed. This systematic review will not help the investigators to establish a causal link between HDP and childhood asthma, but rather to clarify the magnitude of this association.
Response: We appreciate your suggestion. The aim of this systematic review was reframed. Please see lines 86~88 on page 7.

Methods:

Page 8, line 56:
How were case series and case reports handled?
Response: We did not include case series or case reports because these types of studies might not be applicable for an analysis of systematic reviews. To clarify this issue, case series and case reports were listed in the exclusion criteria. Please see line 120 on page 9.

Page 9, line 18:
Clarify what is meant by “studies with overlapping data” and “studies without raw data”.
Response: “Studies with overlapping data” was intended to indicate that the same data from the same population were used in two or more studies. “Studies without raw data” was intended to indicate that no data were provided for a meta-analysis in the studies. Please see lines 122~123 on page 9.

Page 9, line 36:
Do you mean that studies were retrieved from each “article”? (currently written as "database")
Response: We apologize for our error. The correct term should be “article”. We corrected this error. Please see line 127 on page 9.

Page 10, line15:
How did you score the other study designs?
Response: The quality assessment followed our published protocol. The quality of cohort and case-control studies would have been assessed using the Newcastle-Ottawa Scale. The quality of cross-sectional studies would have been assessed using the Agency for Healthcare Research and Quality-recommended 11-item scale. However, the studies finally included in the meta-analysis were all cohort studies; therefore, the other study designs were not included. Please see line 175 on page 12.

Page 10, line 56:
Were all subgroup analyses defined a priori? Were some defined a posteriori? Please clarify this, as well as the rationale for the subgroup analyses. In particular, the rationale for the sibling analysis should be explained here and not in the results section.
Response: We appreciate your comments. A protocol-based subgroup analysis was performed. Because these factors are potential confounding factors that cause heterogeneity, we performed a

subgroup analysis. Because the family situation and genetic factors are similar in siblings, we performed a subgroup analysis of sibling-design studies. This information was added in the methods section. Please see lines 151~154 on page 11.

There is a section missing on the description of how the authors assessed what were “appropriate confounders” adjusted for in each individual study.
I defer to a statistical expert regarding the appropriateness of the analysis used.
Response: Because the approaches used to control confounders substantially differed across the studies, we included the most adjusted and most common variables in the subgroup analysis.

Results:
Clear description of the results.
Response: We revised the description of the results.

Discussion:
Page 19:
The discussion should start with a brief summary of the study findings, without repeating the results. In addition, the overall tone of the discussion should be brought down, as the association found was quite moderate, and I suspect that many confounders were not captured by individual studies.
Response: Thank you for the suggestion. We revised the discussion. Please see lines 261~275 on page 20 and throughout the discussion section. In addition, the overall tone of the discussion was reduced.

Elements currently included in paragraph 1 of the discussion should be featured in the investigator’s perceived strengths and weaknesses section, and do not belong to the first paragraph of the discussion.
Response: We appreciate your suggestion. We moved paragraph 1 of the Discussion to the “Strengths and limitations” section in the Discussion. Please see lines 286~293 on pages 21-22.

Regardless of the causal link between HDP and asthma, HDP appears to be an interesting predictor of asthma in the offspring. Hence, a paragraph in the discussion should expand further on what “early surveillance and interventions” would be warranted for infants of mothers of HDP. Perhaps, assessing whether maternal HDP was present when evaluating infants for asthma would be important.
Response: Thank you for these important suggestions. We added a paragraph regarding the early surveillance of and interventions for infants of mothers of HDP. Please see lines 315~323 on page 23.

Reviewer: 2
Dr. Kawsari Abdullah, Children's Hospital of Eastern Ontario

Introduction:
1. On page 5 line#59-62 authors state some examples of offspring diseases that have been shown to have association with HDP. Here they do not mention respiratory diseases. Mentioning respiratory conditions here can provide a good segue to the next sections where they discuss the association with respiratory diseases in detail.
Response: Dear Dr. Kawsari Abdullah, thank you for your suggestion. We added the respiratory conditions in this section. Please see line 63 on page 5.

Methods:
1. Page 7 line#91: The “data source and search strategy” title heads the wrong paragraph. This paragraph contains information related to the “Eligibility criteria” of the studies. Additionally, these criteria are not complete and lack clarity. Certain information related to the outcome have been provided in the section titled “study selection”. Authors have stated that they have used the PRISMA

checklist for developing the chronology and content of the methods section. However, when I compare the section heading and content provided within these headings to the PRISMA checklist, there are inconsistency and absence of information. One example has been mentioned in my comment above. Hence, I find the methods section of this review messy and lacking the systematic approach of a review.
Response: We appreciate your valuable advice. We followed a preprepared protocol when performing the systematic review [1]. In addition, we made some modifications to the description to enhance clarity. Please see lines 93~123 on pages 7~9.

Ref:
[1] Li P, Xiong T, Hu Y. Hypertensive disorders of pregnancy and risk of asthma in offspring: protocol for a systematic review and meta-analysis. BMJ open 2020;10:e035145.

2. Some secondary outcomes have been stated on page 7 line#95, these include active asthma and asthma exacerbation. Why are these two not considered as asthma? How were they defined?
Response: We appreciate your thorough review. The definition of active asthma includes speaking to a doctor or other health care professional about asthma or taking asthma medication during the past 12 months (International Classification of Diseases, 9th Revision, diagnostic code 493 or International Statistical Classification of Diseases, 10th Revision, diagnostic codes J45 or J46). Asthma exacerbations were defined as a deterioration in asthma requiring any of the following: treatment with an oral corticosteroid, visit to an emergency department, outpatient visit, or hospitalization. Indeed, active asthma and asthma exacerbation represent different asthma statuses.
Because active asthma and asthma exacerbation were considered asthma with different degrees of severity or manifestations, active asthma and asthma exacerbation were set as secondary outcomes in the systematic review. However, only a few included studies investigated active asthma and asthma exacerbations, and we failed to obtain pooled results with regard to the risk of disease severity. Please see lines 114~117 on page 8.

3. On page 9 line#119: What does “literature that was peer reviewed” mean? If the authors only included studies that were published after peer review, it is better to refer to it as “peer reviewed published study” instead of the above.
Response: Thank you for your suggestion. Only peer-reviewed published studies were included in the systematic review. We made a revision. Please see line 119~120 on page 9.

4. Page 10 line#143-144, not clear what authors are meaning by stating - “hazard ratios, incidence rates and RRs were directly considered ORs in the pooled analysis. These effect estimates provide different estimates of risk association. Hence, further clarification on how they were pooled is needed.
Response: Thank you for noting our inaccuracy. These effect estimates have different definitions. The studies included in the systematic review were all cohort studies; however, the results were expressed as either the OR or RR. We referred to a similar systematic review and meta-analysis [1-3] and consulted some statistical experts. We learned that the OR and RR could be summarized under the condition of having the same expression. Therefore, we computed a summary OR and its 95% CIs using the study-specific most-adjusted OR or RR (maternal HDP exposure compared with maternal non-HDP exposure) and the 95% CIs. The hazard ratios (HRs), which were listed in only the study by Mirzakhani et al, represented the risk estimate of mothers with asthma with preeclampsia compared with mothers without asthma without preeclampsia. Thus, we only included the adjusted RR of 1.62 (1.02-2.57) rather than the adjusted HR of 2.68 (1.30-5.61) in the pooled analysis. In addition, the incidence rates should not be included in a pooled analysis, and we apologize for the writing error. We revised the manuscript accordingly. Please see line146~148 on page 10.

Ref:

[1] Chi Z, Qin Y Y , Xin W , et al. Tea consumption and risk of cardiovascular outcomes and total mortality: a systematic review and meta-analysis of prospective observational studies[J]. European Journal of Epidemiology, 2014, 30(2):1-11.
[2[Davies G A, Strader C, Chibbar R, et al. The relationship between physical activity and lymphoma: a systematic review and meta analysis[J]. BMC cancer, 2020, 20(1): 1-16.
[3] Yang W S, Va P, Wong M Y, et al. Soy intake is associated with lower lung cancer risk: results from a meta-analysis of epidemiologic studies[J]. The American journal of clinical nutrition, 2011, 94(6): 1575-1583.

Results:
1. On page 14 line# 180: What does 252,450 cases mean? How are cases defined?
Response: This indicates that 252,450 offspring with asthma were enrolled in the included studies. To enhance clarity, we revised the manuscript. Please see line 191 on page 15.

2. Page 15 line#201: Authors have shown a pooled estimate of the adjusted OR (1.19,95%CI 1.12 – 1.26). However, it is also shown that this estimate had statistically significant heterogeneity (I2 = 72.5%). With this high level of heterogeneity should the authors report the pooled value as their primary estimate of the association between HDP and offspring asthma as depicted in the abstract, results and discussion section of the review.
Response: Thank you very much for your valuable comments.
We noticed that the heterogeneity was significantly high in this systematic review. First, to explore the source of heterogeneity, we performed a subgroup analysis by subgroups of HDP, study quality, study location, offspring ages, singleton births, exposure assessment, outcome assessment, and some adjusted factors. The heterogeneity did not decrease.
When heterogeneity cannot be readily explained, one analytical approach is to incorporate it into a random-effects model [1]. We referred to another similar systematic review [2-4], and a random-effects model was used when I2>50%. The pooled value was used as the primary estimate. Thus, we used a random-effects model.
Finally, we toned down the description of the primary results in this systematic review and elaborated in the limitation section.

Ref:
[1] Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for systematic reviews of interventions. Type]: Chichester (UK: John Wiley & Sons, 2019.
[2]Maher GM, O'Keeffe GW, Kearney PM,et al. Association of Hypertensive Disorders of Pregnancy With Risk of Neurodevelopmental Disorders in Offspring: A Systematic Review and Meta-analysis. JAMA Psychiatry. 2018;75(8):809-819.
[3] Tusting L S, Willey B, Lucas H, et al. Socioeconomic development as an intervention against malaria: a systematic review and meta-analysis[J]. The Lancet, 2013, 382(9896): 963-972.
[4] Cho Y A, Kim J. Effect of probiotics on blood lipid concentrations: a meta-analysis of randomized controlled trials[J]. Medicine, 2015, 94(43).

3. Subgroup analysis results: Within each of the subgroup analysis (table 1), the ORs shown as statistically significant, also show high level of heterogeneity (category of HDP= 77%; sibling design (no) = 75.8%; study quality = 66.9%; location = 78.5%; exposure assessment = 70.1%; outcome assessment = 82.8%). Based on this observation, the question again arises whether pooling of the effect estimates (ORs) reported in each of the selected studies is justifiable?
Response: We appreciate your thorough review. We performed a subgroup analysis and found that the heterogeneity cannot be explained by a subgroup analysis. We adopted the Cochrane handbook method as follows: when heterogeneity cannot be readily explained, one analytical approach is to incorporate it into a random-effects model [1]. This method was used in a similar systematic review when there was significant heterogeneity (I2>50%) [2-4].

Ref:
[1] Higgins JPT, Thomas J, Chandler J, et al. Cochrane Handbook for systematic reviews of interventions. Type]: Chichester (UK: John Wiley & Sons, 2019.
[2]Maher GM, O'Keeffe GW, Kearney PM,et al. Association of Hypertensive Disorders of Pregnancy With Risk of Neurodevelopmental Disorders in Offspring: A Systematic Review and Meta-analysis. JAMA Psychiatry. 2018;75(8):809-819.
[3] Tusting L S, Willey B, Lucas H, et al. Socioeconomic development as an intervention against malaria: a systematic review and meta-analysis[J]. The Lancet, 2013, 382(9896): 963-972.
[4] Cho Y A, Kim J. Effect of probiotics on blood lipid concentrations: a meta-analysis of randomized controlled trials[J]. Medicine, 2015, 94(43).

4. Subgroup analysis results: It has been stated on page 18 line#235 that after adjusting for maternal smoking during pregnancy the heterogeneity has decreased. However, in table 1 the heterogeneity reported without adjustment for this variable is less than the adjusted one (adjusted (30.7) vs. not adjusted (29.9)). Please verify why?
Response: The heterogeneity decreased after adjusting for maternal smoking, which was compared with the overall studies (smoking adjusted and not adjusted). After dividing the study into smoking adjusted and smoking not adjusted, the heterogeneity within the two types of studies was reduced compared with the overall heterogeneity.

5. Formatting of table 2:
What does the column heading “heterogeneity statistic” imply?
Having the I2 column before the OR is confusing.
Response:
The “heterogeneity statistic” indicates the statistical value of the heterogeneity test. To avoid confusion, we deleted this column.
The I2 column was moved to the column after OR.

6. Supplemental Table 4: Does each row estimate indicate the pooled ORs after excluding that particular study? If so, a heading or a footnote is needed. Otherwise, the results given in this table are not interpretable just on own its own.
Response: Thank you for your suggestion. The estimate shown in each row indicates the pooled ORs after excluding that particular study. We added a footnote under the table. Please see line 107 on page 10 in supplemental table 4.

Discussion and conclusion:
1. Page 20 line#278: there should not be a dash after two.
Response: Thank you. We considered that the "two sibling-design studies" should be used when only two studies with a sibling design were considered in analysis. However, the "two-sibling-design studies" meant the studies investigated groups of two siblings (pairs). We thought the latter might be more appropriate in this systematic review.

2. Page 21 line#292: Authors have stated that misclassification due to different assessment of HDP and Outcomes. From table 1 it is evident that many of the studies used ICD codes for ascertainment of HDP. A subgroup analysis has been done based on exposure assessment where the categories were maternal-report and medical records. Should this category be clinical diagnosis of HDP vs. ICD codes based medical records? Clinical method is the gold standard for any disease diagnosis and able to provide a more robust evidence associated with any risk association. This fact may need to be highlighted in the conclusion where the authors are calling for further research.

Response: Thank you for your suggestion. We thought that maternal-reported asthma should be based on previous medical records. We highlighted “using the gold standard for disease diagnosis” in the conclusion. Please see lines 328~329 on page 25.

Reviewer: 3
Prof. Zhengping Liu, Southern Medical University

1. Complete the result of records obtained by each search strategy in Supplemental eTable 1.
Response: Dear Prof. Zhengping Liu, thank you for this suggestion. We supplemented the search records in each search strategy.

2. The authors should explain the rationale for the statistical method of combining ORs for subtypes of HDPs. The definitions of exposures (GH, PE and HbP) are very different though they are subtypes of HDPs. Otherwise, the meta-analysis should be performed by subtypes separately.
Response: We appreciate your suggestion. The aim of this systematic review was to summarize the available evidence regarding the association between HDP and the risk of asthma in offspring. According to the International Society of Hypertension in Pregnancy, HDP were defined as any hypertension condition (systolic blood pressure (BP) ≥140 and/or diastolic BP ≥90 mmHg) manifesting before gestation or before 20 weeks and hypertension starting at or after 20 weeks. GH, PE and HbP are all subtypes of HDP, although they have very different characteristics. Therefore, we summarized the pooled estimate and performed a subgroup analysis by categorizing PE and other HDPs. This method was also applied in a similar systematic review [1]. The detailed results are shown in Supplemental eFigure 1 Forest plot of studies investigating the association between subtypes of HDP and asthma in offspring.

Ref:
[1] Maher GM, O'Keeffe GW, Kearney PM,et al. Association of Hypertensive Disorders of Pregnancy With Risk of Neurodevelopmental Disorders in Offspring: A Systematic Review and Meta-analysis. JAMA Psychiatry. 2018;75(8):809-819.

3. For the study by Byberg et al., the ORs for MM-PE and S-PE were derived from the same logistic model, in which no-PE was the shared reference. Therefore, directly combining these ORs in the meta-analysis is problematic since these ORs are not independent. Alternatively, a fixed effects model is suggested to calculate a combined OR for PE and this combined OR can be used for meta-analysis afterwards. [1]
[1] Dong JY, Zhang YH, Qin LQ. Erectile dysfunction and risk of cardiovascular disease: meta-analysis of prospective cohort studies. J Am Coll Cardiol. 2011 Sep 20;58(13):1378-85. doi: 10.1016/j.jacc.2011.06.024. PMID: 21920268.
Response: Thank you very much for your suggestion. We referred to another meta-analysis and found that the ORs derived from the same logistic model were directly combined using a random-effects model [1]. In addition, the I2 statistic (significance level, >50%) was applied to examine heterogeneity. In this systematic review, statistically significant heterogeneity with I2 = 72.5% was found. Thus, a random-effects model should be applied to calculate the pooled ORs.

Ref:
[1] Maher GM, O'Keeffe GW, Kearney PM,et al. Association of Hypertensive Disorders of Pregnancy With Risk of Neurodevelopmental Disorders in Offspring: A Systematic Review and Meta-analysis. JAMA Psychiatry. 2018;75(8):809-819.

VERSION 2 – REVIEW

REVIEWER	Liu, Zhengping Southern Medical University, Department of Obsterics
REVIEW RETURNED	05-Jul-2021

GENERAL COMMENTS	The authors explained clearly regarding to the questions and make a good revisions according to the suggestions. I think that the revised manuscript can be accepted for publication.